

1 Solution — GIAM 7.1

Problem 9

1.1 Problem

How many handshakes will transpire when n people first meet?

1.2 Setup

Let P be the set of the n people present. Model “first meeting” by the following two conventions:

- **Every pair meets.** Every two distinct people $p, q \in P$ shake hands.
- **Exactly once.** Each pair shakes hands a single time, and a handshake is unordered: p shaking q ’s hand is the same event as q shaking p ’s.

Under these conventions a *handshake* is precisely an unordered pair $\{p, q\} \subseteq P$ of distinct people, and the question reduces to counting such pairs.

1.3 Claim

$$\boxed{\# \text{handshakes} = \binom{n}{2} = \frac{n(n-1)}{2}.}$$

We give three proofs.

1.4 Proof 1 — Counting Pairs Directly

The number of 2-element subsets of an n -element set is, by definition,

$$\binom{n}{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}. \quad \blacksquare$$

1.5 Proof 2 — Double Counting

Count ordered pairs $(p, \{p, q\})$, where p is a person and $\{p, q\}$ is a handshake p participates in.

- *Sum over people.* Each of the n people participates in exactly $n - 1$ handshakes (one with each other person), giving $n(n - 1)$ ordered pairs.
- *Sum over handshakes.* Each handshake $\{p, q\}$ contributes exactly 2 ordered pairs — one for p and one for q .

Equating the two counts:

$$n(n - 1) = 2 \cdot \# \text{handshakes} \implies \# \text{handshakes} = \frac{n(n - 1)}{2}. \quad \blacksquare$$

1.6 Proof 3 — Summing by Arrival Order

Order the people $p_1, p_2, \dots, p_n$ by who walks in first. When p_i arrives, the $i - 1$ people already present each shake p_i 's hand — contributing $i - 1$ *new* handshakes. (No other handshakes among the earlier arrivals are introduced; those already happened.) Summing over i ,

$$\# \text{handshakes} = \sum_{i=1}^n (i - 1) = 0 + 1 + 2 + \dots + (n - 1) = \frac{(n - 1)n}{2},$$

using the standard arithmetic-progression sum $\sum_{k=0}^m k = m(m + 1)/2$. $\blacksquare$

1.7 Sanity Check on Small n

n	People	Handshakes (enumerated)	$\binom{n}{2} = \frac{n(n - 1)}{2}$
1	$\{A\}$	(none)	0
2	$\{A, B\}$	AB	1
3	$\{A, B, C\}$	AB, AC, BC	3
4	$\{A, B, C, D\}$	AB, AC, AD, BC, BD, CD	6

5	$\{A, B, C, D, E\}$	(as above) + AE, BE, CE, DE	10
6	$\{A, B, C, D, E, F\}$	(as above) + AF, BF, CF, DF, EF	15

Table 1.1

Each new arrival adds a triangle's worth of new handshakes, illustrating Proof 3.

1.8 Remark — Triangular Numbers

The sequence

$$0, 0, 1, 3, 6, 10, 15, 21, 28, \dots$$

obtained by tabulating $\binom{n}{2}$ for $n = 0, 1, 2, \dots$ is precisely the sequence of *triangular numbers* (shifted). Recalling $T_m = m(m+1)/2$ from Chapter 6.6,

$$\binom{n}{2} = \frac{n(n-1)}{2} = T_{n-1}.$$

So the answer to the handshake problem for n people is the $(n-1)$ -th triangular number — a clean alternative phrasing of the same count.

1.9 Remark — Graph-Theoretic Reading

Identify each person with a vertex and each handshake with an edge between the two participants. The resulting graph has every possible edge present, i.e., it is the *complete graph* K_n . The number of edges of K_n is exactly the handshake count,

$$|E(K_n)| = \binom{n}{2} = \frac{n(n-1)}{2}.$$

This perspective is useful in §7.2 (parity and counting arguments), where the *handshaking lemma* states that the sum of vertex degrees of any graph equals twice the number of edges — exactly the double-counting identity used in Proof 2.